🎫 Ticket Creation

All Cases Analysis

8 Cases | 15+ Flowcharts | Complete Testing Guide

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📋 Tổng Quan

Phân tích chi tiết **TẤT CẢ** các trường hợp (cases) khi người dùng tạo ticket trong hệ thống IT Ticket Management, từ flow đơn giản đến phức tạp, kèm sơ đồ, SQL, code examples và testing checklist.

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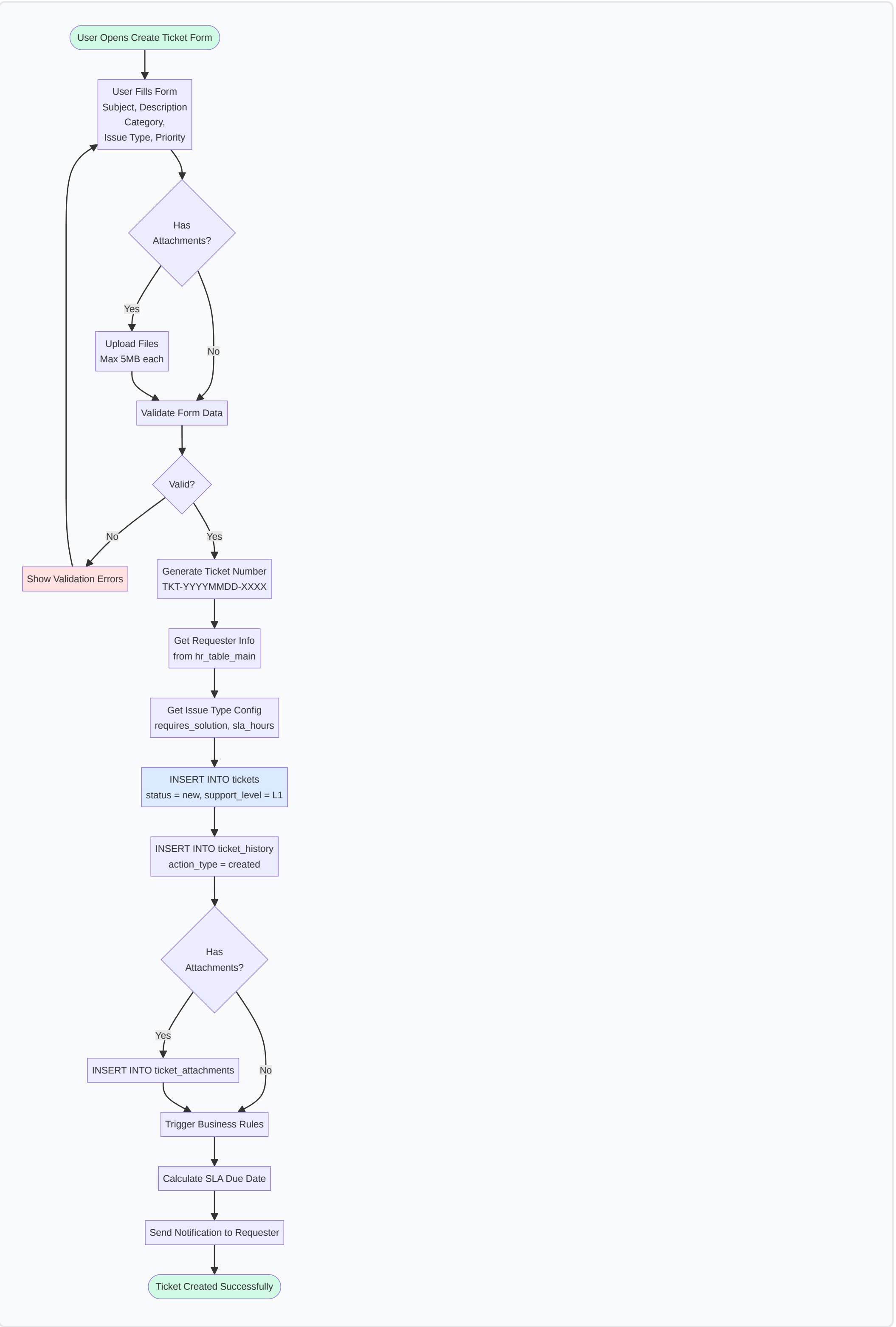
Case 1 Normal Ticket Creation

📝 **Description:** User tạo ticket cơ bản với đầy đủ thông tin bắt buộc (subject, description, category, issue type, priority).

🎯 Flow Diagram



User Opens Create Ticket Form
User Fills Form Subject, Description Category, Issue Type, Priority
Has Attachments?
Yes
Upload Files Max 5MB each
No
Validate Form Data
Valid?
No
Show Validation Errors
Yes
Generate Ticket Number TKT-YYYYMMDD-XXXX
Get Requester Info from hr_table_main
Get Issue Type Config requires_solution, sla_hours
INSERT INTO tickets status = new, support_level = L1
INSERT INTO ticket_history action_type = created
Has Attachments?
Yes
INSERT INTO ticket_attachments
No
Trigger Business Rules
Calculate SLA Due Date
Send Notification to Requester
Ticket Created Successfully

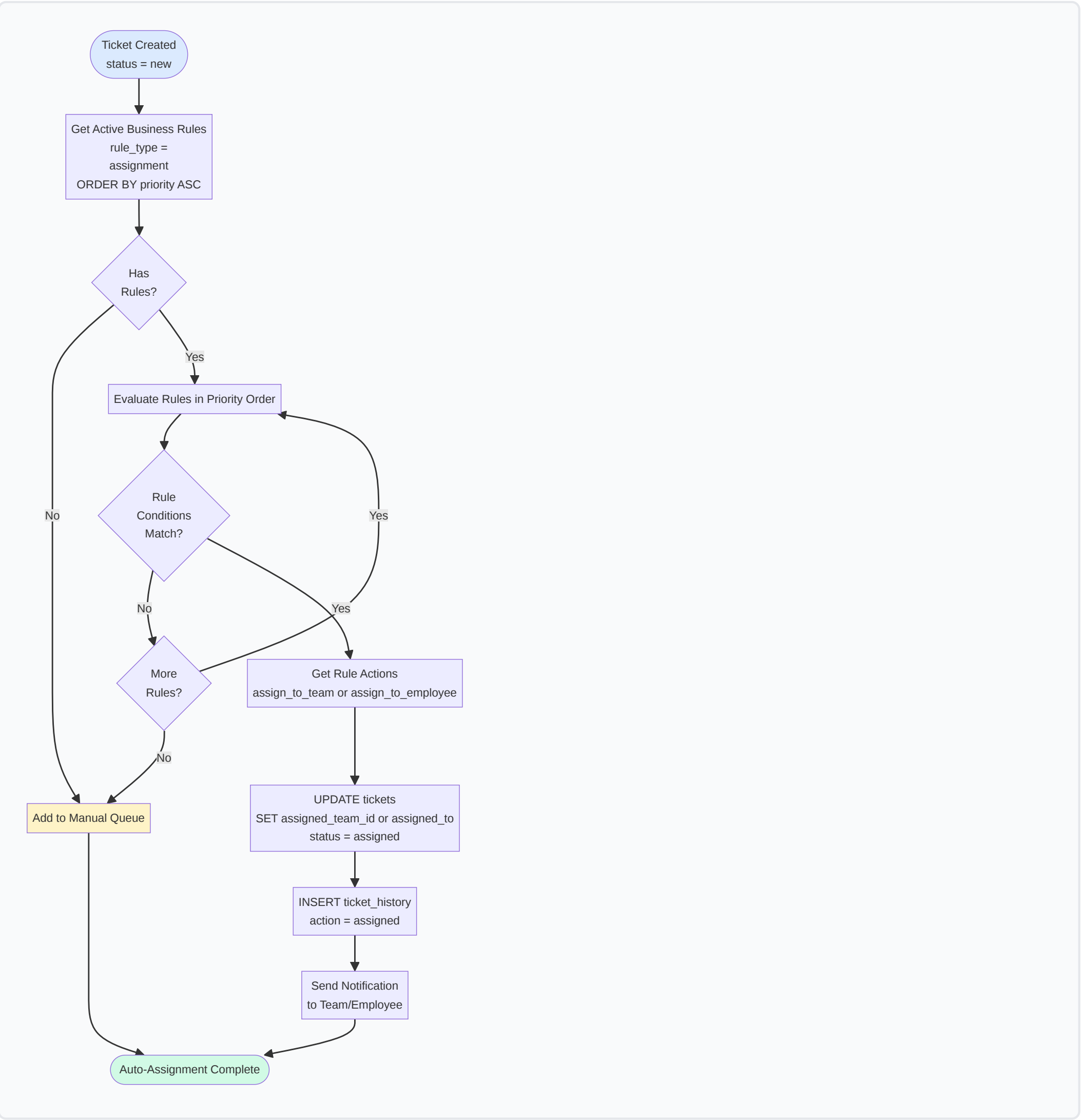
✅ Success Criteria

- ✅ Ticket number generated uniquely
- ✅ Ticket saved to database
- ✅ History logged
- ✅ Attachments uploaded (if any)
- ✅ Requester notified

Case 2 Auto-Assignment by Business Rules

📝 **Description:** Sau khi ticket được tạo, hệ thống tự động assign ticket cho team/tech dựa trên business rules (category, priority, department, etc.).

🎯 Flow Diagram



Ticket Created status = new
Get Active Business Rules rule_type = assignment ORDER BY priority ASC
Has Rules?
Yes
No
Evaluate Rules in Priority Order
Rule Conditions Match?
No
Yes
More Rules?
Yes
No
Add to Manual Queue
Get Rule Actions assign_to_team or assign_to_employee
UPDATE tickets SET assigned_team_id or assigned_to status = assigned
INSERT ticket_history action = assigned
Send Notification to Team/Employee
Auto-Assignment Complete

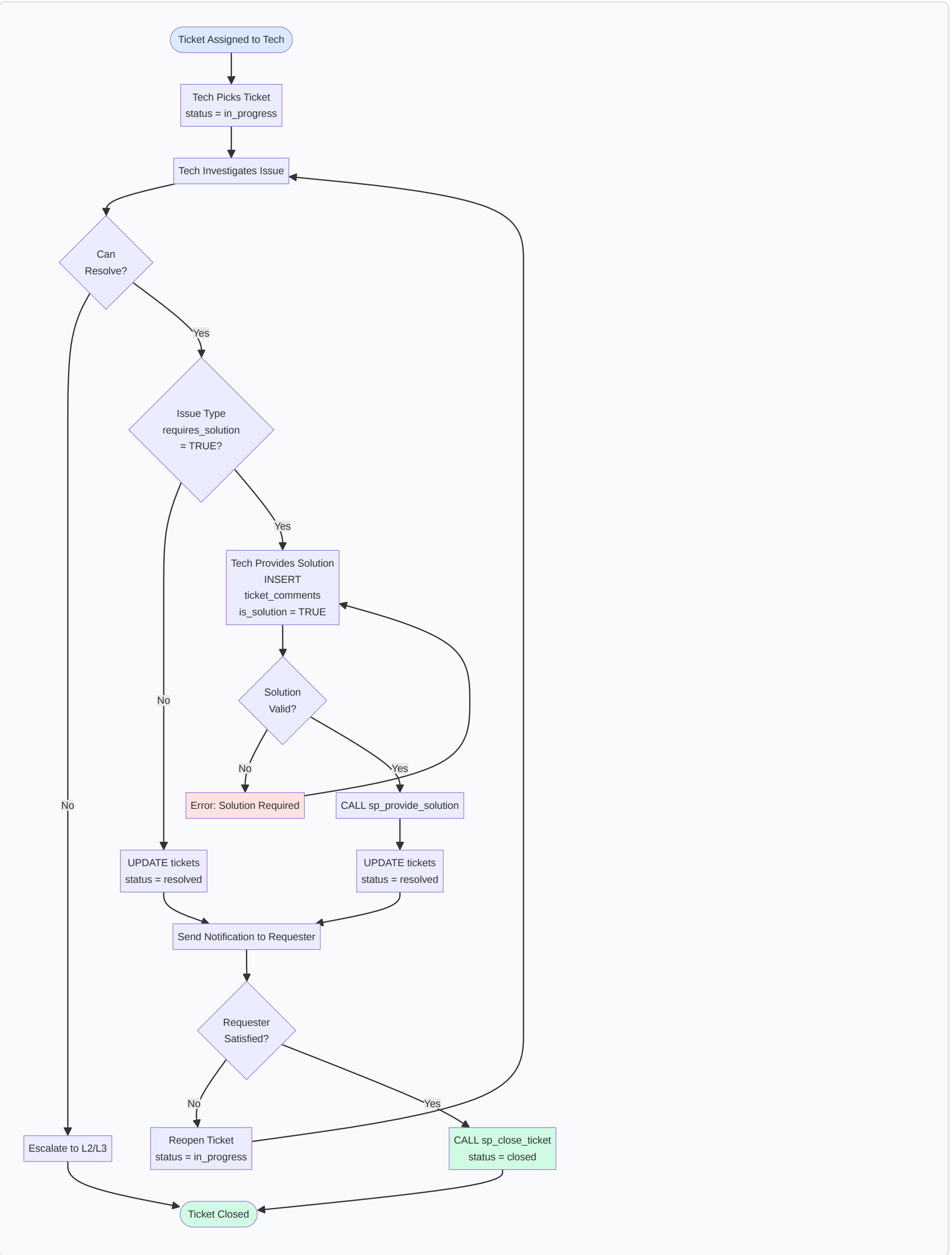
📊 Business Rules Examples

```
// Rule 1: Network issues → Network Team
{
  "name": "Network Issues to Network Team",
  "rule_type": "assignment",
  "priority": 1,
  "conditions": {
    "category_code": "NETWORK",
    "priority": ["high", "critical"]
  },
  "actions": {
    "assign_to_team": 1,
    "set_support_level": "L1"
  }
}
```

Case 3 Ticket Requires Solution

⚠️ **Important:** Một số issue types yêu cầu tech phải provide solution trước khi close ticket. Trigger và stored procedure sẽ validate điều này.

🎯 Flow Diagram

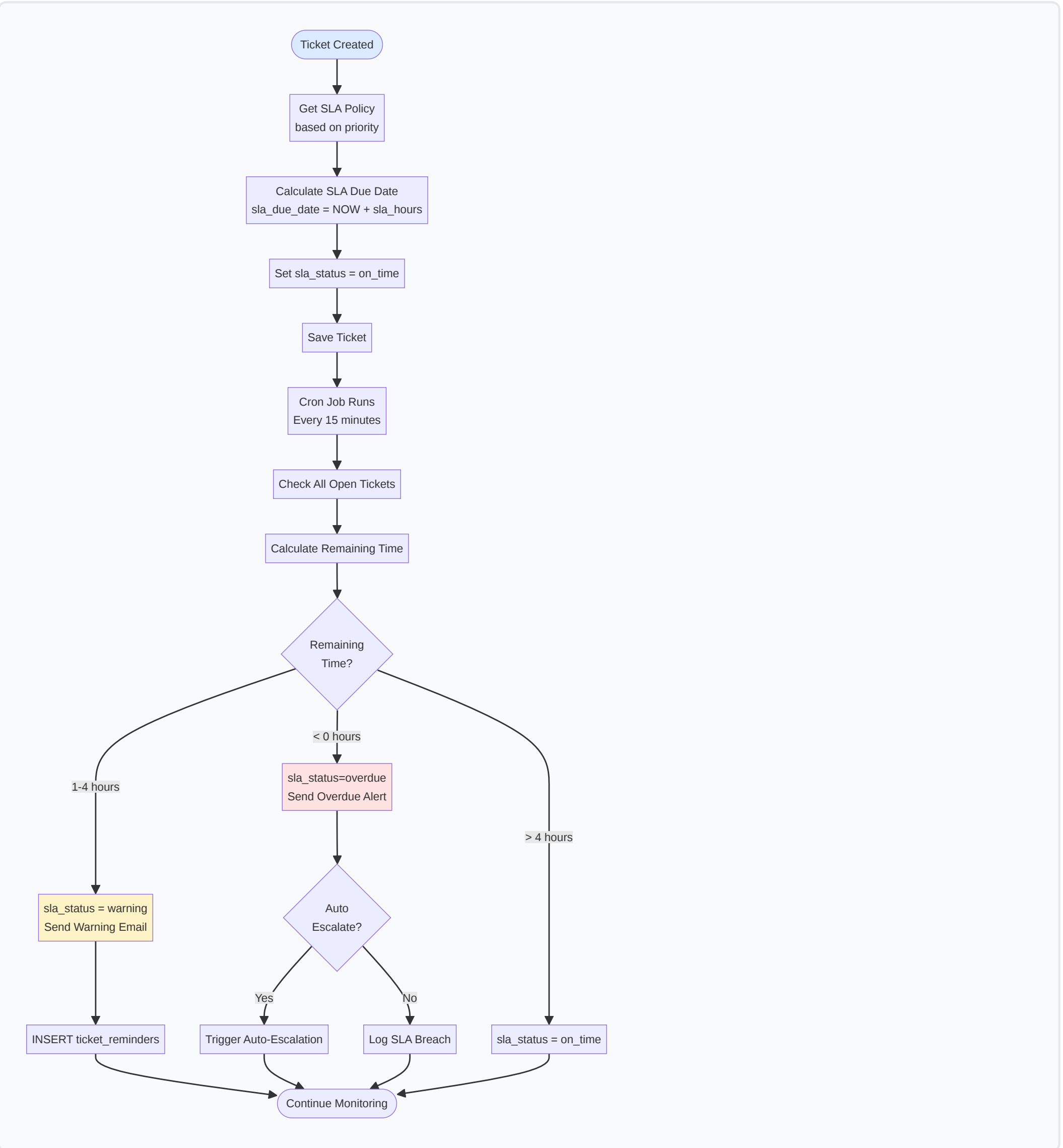


Ticket Assigned to Tech
Tech Picks Ticket status = in_progress
Tech Investigates Issue
Can Resolve?
Yes
Issue Type requires_solution = TRUE?
Yes
Tech Provides Solution INSERT ticket_comments is_solution = TRUE
Solution Valid?
No
Error: Solution Required
Yes
CALL sp_provide_solution
No
UPDATE tickets status = resolved
UPDATE tickets status = resolved
Send Notification to Requester
Requester Satisfied?
No
Reopen Ticket status = in_progress
Yes
CALL sp_close_ticket status = closed
No
Escalate to L2/L3
Ticket Closed

Case 4 SLA Policy Application

📝 **Description:** Mỗi ticket có SLA deadline dựa trên priority. Hệ thống track và warning khi gần quá hạn (1-4 hours), overdue khi quá hạn.

🎯 Flow Diagram



Ticket Created
Get SLA Policy based on priority
Calculate SLA Due Date sla_due_date = NOW + sla_hours
Set sla_status = on_time
Save Ticket
Cron Job Runs Every 15 minutes
Check All Open Tickets
Calculate Remaining Time
Remaining Time?
1-4 hours
< 0 hours
> 4 hours
sla_status=overdue Send Overdue Alert
sla_status = warning Send Warning Email
Auto Escalate?
Yes
No
INSERT ticket_reminders
Trigger Auto-Escalation
Log SLA Breach
sla_status = on_time
Continue Monitoring

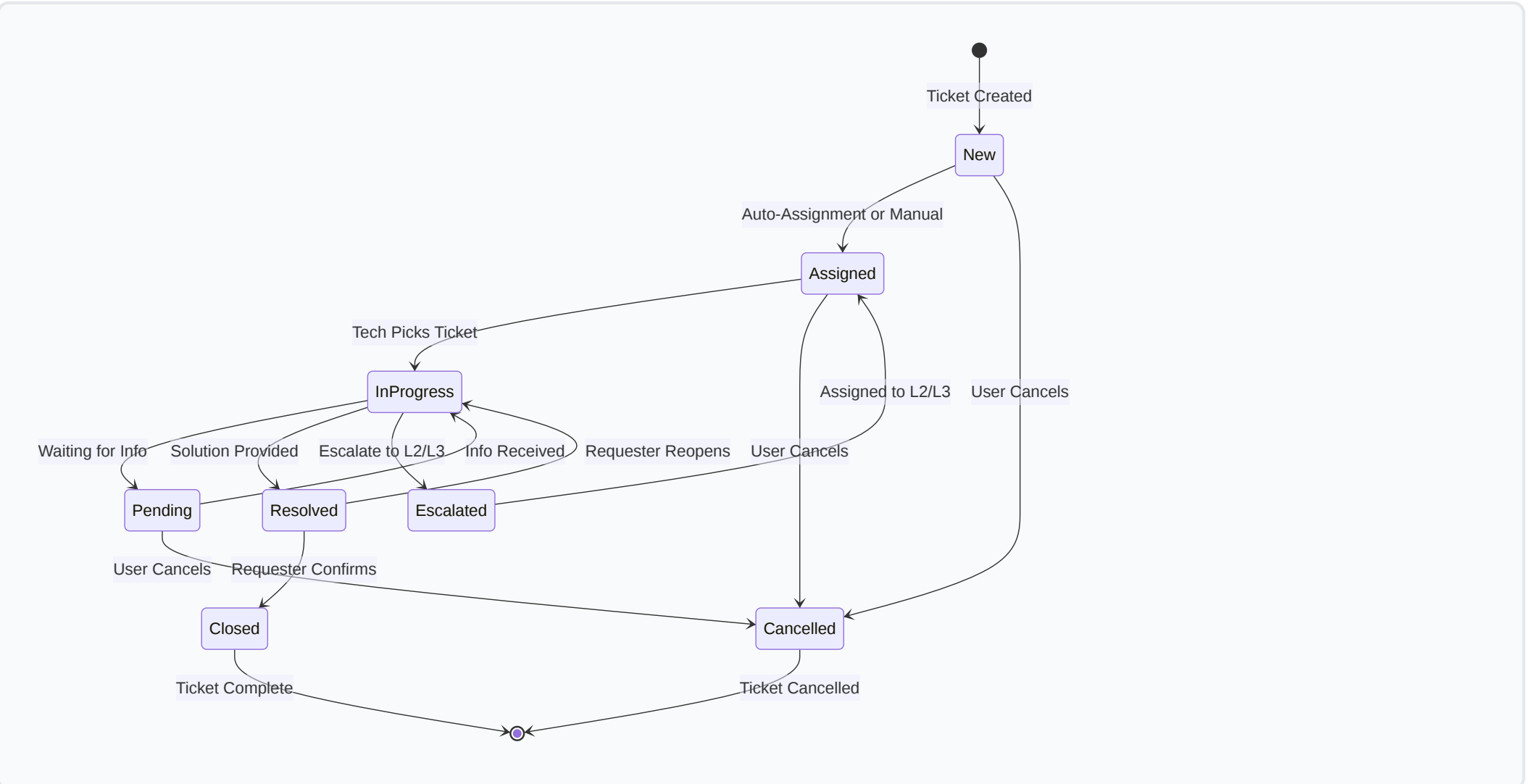
📊 SLA Policies

Priority	Response Hours	Resolution Hours	Warning Threshold
Critical	1 hour	4 hours	1-4 hours remaining
High	2 hours	8 hours	1-4 hours remaining
Medium	4 hours	24 hours	1-4 hours remaining
Low	8 hours	72 hours	1-4 hours remaining

Case 5 Workflow State Machine

📝 **Description:** Ticket theo workflow với các states và transitions được định nghĩa trước. Mỗi transition có conditions (roles, requirements).

🎯 State Diagram

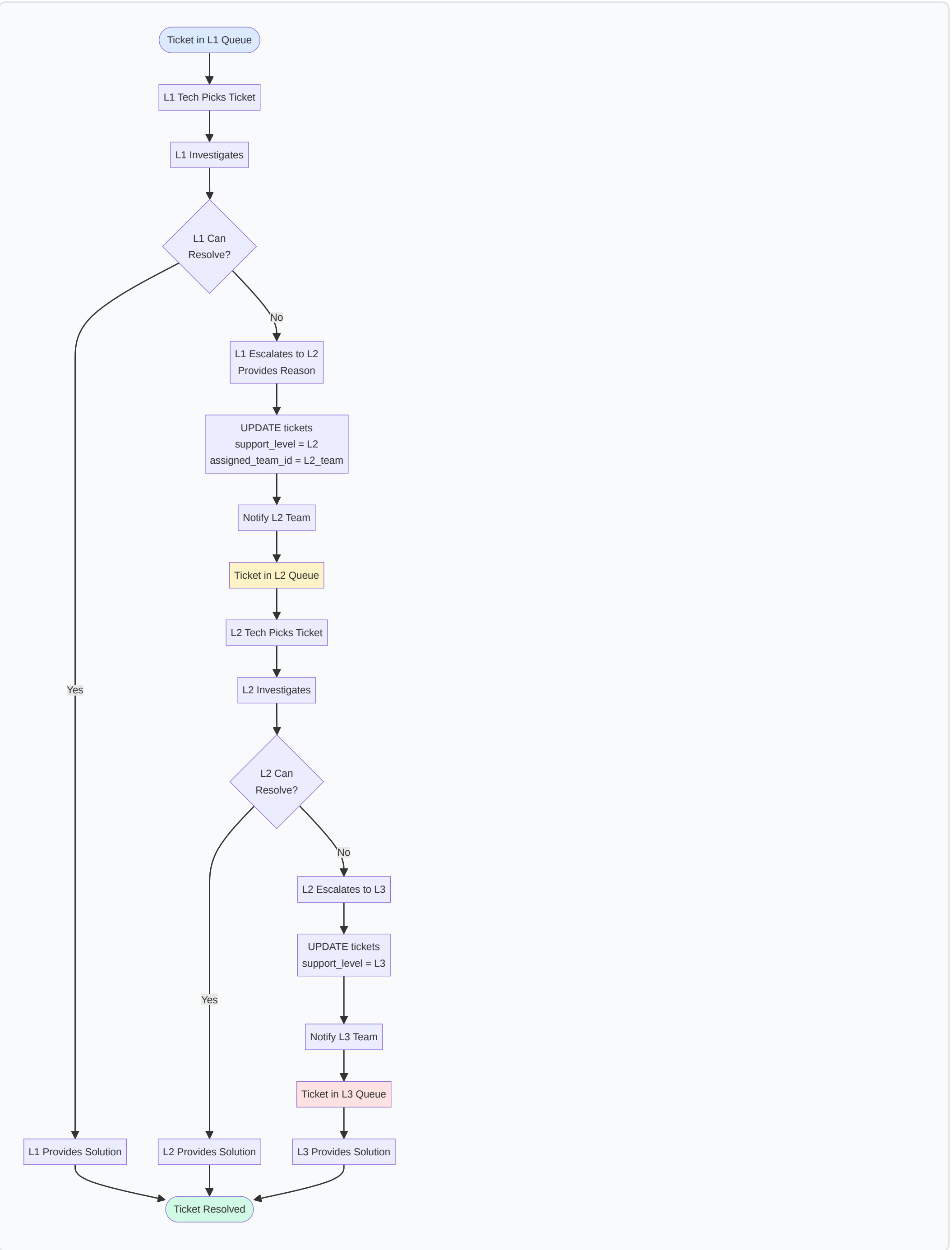


Ticket Created
New
Auto-Assignment or Manual
Assigned
Tech Picks Ticket
InProgress
Assigned to L2/L3
User Cancels
Waiting for Info
Solution Provided
Escalate to L2/L3
Info Received
Requester Reopens
User Cancels
Pending
Resolved
Escalated
User Cancels
Requester Confirms
Closed
Cancelled
Ticket Complete
Ticket Cancelled

Case 6 Multi-Level Escalation

⚠️ **Escalation Path:** L1 → L2 → L3. Mỗi escalation cần reason và được log vào ticket_escalations.

🎯 Flow Diagram



Ticket in L1 Queue
L1 Tech Picks Ticket
L1 Investigates
L1 Can Resolve?
No
L1 Escalates to L2 Provides Reason
UPDATE tickets support_level = L2 assigned_team_id = L2_team
Notify L2 Team
Ticket in L2 Queue
L2 Tech Picks Ticket
L2 Investigates
L2 Can Resolve?
No
L2 Escalates to L3
UPDATE tickets support_level = L3
Notify L3 Team
Ticket in L3 Queue
Yes
Yes
L1 Provides Solution
L2 Provides Solution
L3 Provides Solution
Ticket Resolved

Case 7 Notification Triggers

📝 **Description:** Tất cả các events trigger notifications cho các stakeholders liên quan (requester, assigned tech, manager, team).

📊 Notification Matrix

Event	Recipient	Notification Type	Priority
Ticket Created	Requester	Confirmation	Normal
Ticket Assigned	Assigned Tech	New Assignment	Normal
Solution Provided	Requester	Solution Ready	High
SLA Warning	Assigned Tech + Manager	SLA Alert	High
SLA Overdue	Manager + Team Lead	SLA Breach	Critical
Escalated	L2/L3 Team + Manager	Escalation Alert	High
Ticket Closed	Requester + Assigned Tech	Closure Confirmation	Normal

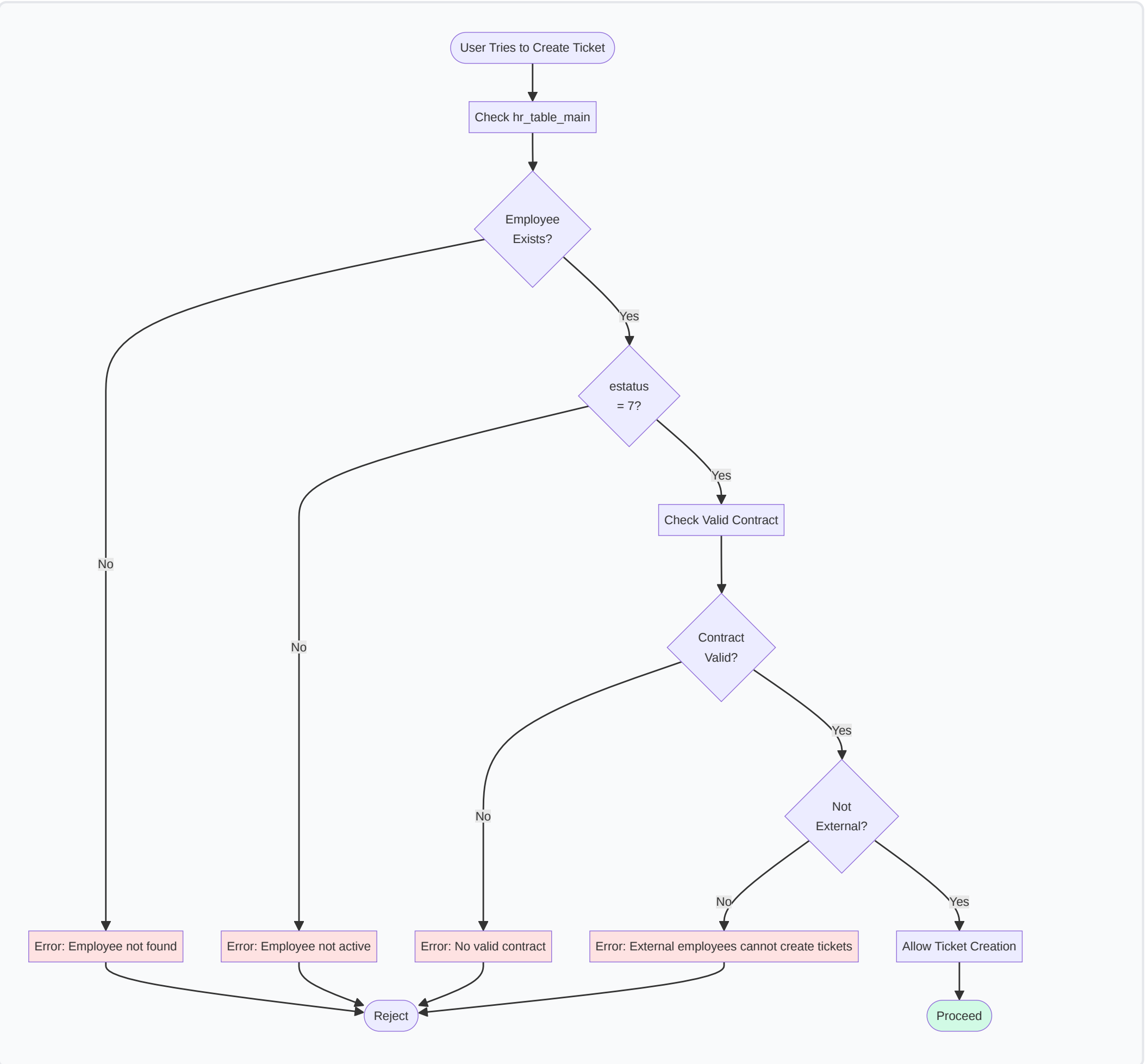
Case 8 Edge Cases & Validations

🚨 **Critical Validations:** Các edge cases phải được handle gracefully với user-friendly error messages.

📋 Edge Cases List

1. **Requester Not Active Employee** - Check estatus = 7, valid contract, not external
2. **Duplicate Ticket Detection** - Same requester, similar subject, within 24 hours
3. **Category/Issue Type Inactive** - Validate status = 'active'
4. **File Upload Validation** - Max 5MB, allowed types, malware scan
5. **Concurrent Updates** - Optimistic locking with updated_at
6. **Ticket Number Collision** - Use transaction + FOR UPDATE lock
7. **SLA Calculation Edge Cases** - Skip weekends, holidays, non-business hours

🎯 Validation Flow



User Tries to Create Ticket
Check hr_table_main
Employee Exists?
Yes
estatus = 7?
Yes
Check Valid Contract
Contract Valid?
Yes
Not External?
No
No
No
No
Yes
Error: Employee not found
Error: Employee not active
Error: No valid contract
Error: External employees cannot create tickets
Allow Ticket Creation
Reject
Proceed

🎯 Complete Testing Checklist

Case 1: Normal Creation

☐ Create ticket with all required fields
☐ Create ticket with attachments
☐ Verify ticket number generation
☐ Verify requester notification
☐ Verify history logging

Case 2: Auto-Assignment

☐ Test rule matching by category
☐ Test rule matching by priority
☐ Test rule matching by department
☐ Test multiple rules (priority order)
☐ Test no matching rules (manual queue)

Case 3: Requires Solution

☐ Try to resolve without solution (should fail)
☐ Provide solution and resolve
☐ Try to close without solution (should fail)
☐ Verify solution in ticket_comments

Case 4: SLA Tracking

☐ Verify SLA due date calculation
☐ Test SLA warning (1-4 hours)
☐ Test SLA overdue (past deadline)
☐ Verify email notifications
☐ Test SLA with weekends/holidays

Case 5: Workflow

☐ Test all valid transitions
☐ Test invalid transitions (should fail)
☐ Test role-based transitions
☐ Test condition-based transitions

Case 6: Escalation

☐ Escalate L1 → L2
☐ Escalate L2 → L3
☐ Verify team reassignment
☐ Verify notifications
☐ Verify escalation history

Case 7: Notifications

☐ Test all notification types
☐ Verify in-app notifications
☐ Verify email queue
☐ Test notification preferences

Case 8: Edge Cases

☐ Inactive employee creates ticket
☐ Duplicate ticket detection
☐ Inactive category/issue type
☐ Invalid file upload
☐ Concurrent updates
☐ Ticket number collision

📝 Notes

- All cases are interconnected and can occur simultaneously
- Business rules can trigger multiple actions
- Notifications are asynchronous (queued)
- SLA tracking runs independently via cron
- Workflows enforce state machine logic
- Edge cases must be handled gracefully with user-friendly errors

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